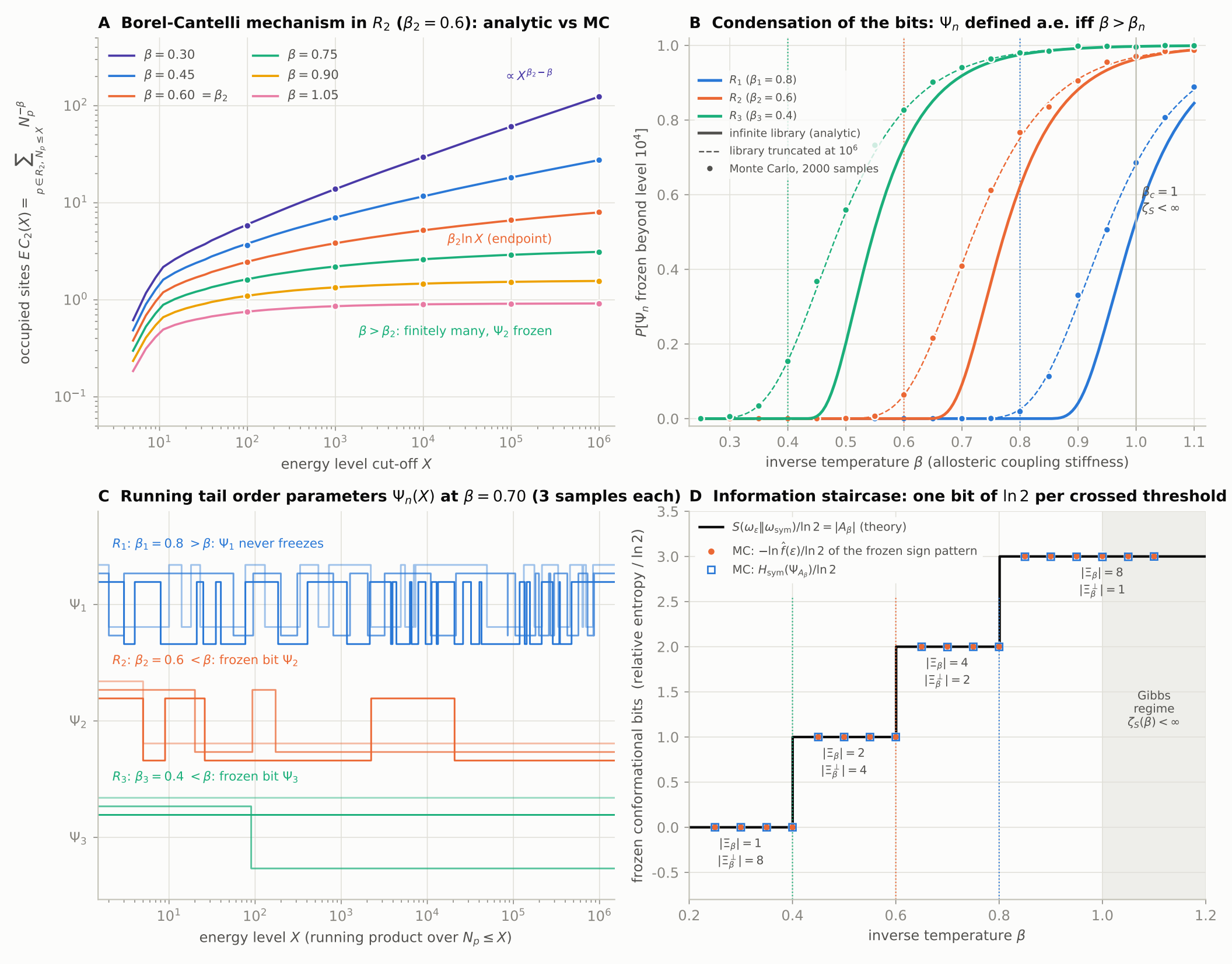

A Borel-Cantelli mechanism in R_2 ($\beta_2 = 0.6$): analytic vs MC
$\beta = 0.30$
$\beta = 0.45$
$\beta = 0.60 = \beta_2$
$\beta = 0.75$
$\beta = 0.90$
$\beta = 1.05$
$\propto X^{\beta_2 - \beta}$
$\beta_2 \ln X$ (endpoint)
$\beta > \beta_2$: finitely many, Ψ_2 frozen
occupied sites $EC_2(X) = \sum_{p \in R_2, N_p \le X} N_p^{-\beta}$
energy level cut-off X
B Condensation of the bits: Ψ_n defined a.e. iff $\beta > \beta_n$
R_1 ($\beta_1 = 0.8$)
R_2 ($\beta_2 = 0.6$)
R_3 ($\beta_3 = 0.4$)
infinite library (analytic)
library truncated at 10^6
Monte Carlo, 2000 samples
$\beta_c = 1$
$\zeta_S < \infty$
$P[\Psi_n$ frozen beyond level $10^4]$
inverse temperature β (allosteric coupling stiffness)
C Running tail order parameters $\Psi_n(X)$ at $\beta = 0.70$ (3 samples each)
R_1: $\beta_1 = 0.8 > \beta$: Ψ_1 never freezes
R_2: $\beta_2 = 0.6 < \beta$: frozen bit Ψ_2
R_3: $\beta_3 = 0.4 < \beta$: frozen bit Ψ_3
Ψ_1
Ψ_2
Ψ_3
energy level X (running product over $N_p \le X$)
D Information staircase: one bit of $\ln 2$ per crossed threshold
$S(\omega_\varepsilon \| \omega_{\mathrm{sym}})/\ln 2 = |A_\beta|$ (theory)
MC: $-\ln \hat{f}(\varepsilon)/\ln 2$ of the frozen sign pattern
MC: $H_{\mathrm{sym}}(\Psi_{A_\beta})/\ln 2$
$|\Xi_\beta| = 1$, $|\Xi_\beta^\perp| = 8$
$|\Xi_\beta| = 2$, $|\Xi_\beta^\perp| = 4$
$|\Xi_\beta| = 4$, $|\Xi_\beta^\perp| = 2$
$|\Xi_\beta| = 8$, $|\Xi_\beta^\perp| = 1$
Gibbs regime $\zeta_S(\beta) < \infty$
frozen conformational bits (relative entropy / $\ln 2$)
inverse temperature β